



DPUDYNCar

The “**DYN**amics” DPUs can be used to simulate the vehicle dynamics within SILAB. The DPUDYNCar represents the dynamics of a passenger car without trailer. This car is modeled as a rigid vehicle body with four independent suspension modules. Each suspension module is connected to a simplified tire model which is based on Pacejka’s Magic Formula. A set of customizable parameter files allows to generate various car setups.

Disclaimer:

The model interacts closely e.g. with mockup components like force feedback steering devices, such that inconsistencies in the respective DPU configurations might cause misbehavior of the overall system. Please contact a WIVW technician for help with setting up your model, if needed.

1. Inputs:

#Simulation control & debugging

<i>ReloadConfiguration</i>	Applies saved changes in the configuration files when set To 1
<i>FirstStart</i>	Disables the ignition/start routine when set to 0
<i>Reset</i>	Resets some integrators and error counters within the model
<i>Debug1...8</i>	Not Connected

#Vehicle States

<i>ABS_on</i>	[0 1] flag to turn ABS functionality on.
<i>Ignition</i>	[0 1] car ignition switch
<i>Starter</i>	[0 1] car starter button
<i>SteeringWheel</i>	[rad] steering wheel angle
<i>AcceleratorPedal</i>	[0...1] accelerator actuation
<i>ClutchPedal</i>	[0...1] accelerator actuation
<i>BrakePedal</i>	[0...1] foot brake actuation
<i>HandBrake</i>	[0...1] hand brake actuation
<i>GearManual</i>	[-1....n] manual gear selection (@ Gbx_mode = 1)
<i>GearAutomatic</i>	-1 : reverse (@ Gbx_mode = 2)
	0 : neutral
	1 : drive
	2 : lock gears above n-2
	3 : lock gears above n-3
	.. : lock gears above n-..
	m : lock gears above n-m



SILAB DRIVING SIMULATION

version Version 7.1 documentation

#Road contact

<i>Zfl, _fr, _rl, _rr</i>	[m] height of front-left (etc.) tire contact point
<i>mufl, _fr, _rl, _rr</i>	[1] friction coefficient gains at each tire contact point
<i>XOffset, YOffset, ZOffset</i>	[m] Additional translation of the vehicle in the simulation's coordinate system during the simulation. This input is computed by DPUSCNX. Default: 0.0 m

#Steering system

<i>FFB_Friction</i>	[Nm] static friction torque resisting steering wheel rotation
<i>FFB_Damping</i>	[Nms/rad] damping torque proportional to steering wheel speed
<i>delta_ctrl</i>	[0 1] activate the steer angle controller
<i>delta_set</i>	[rad] set-point for the steer angle controller
<i>effect_ID</i>	[0,1,2,3] select the effect type 0 = off 1 = sine wave 2 = square wave 3 = saw tooth
<i>effect_freq</i>	[Hz] Frequency of the selected effect
<i>effect_gain</i>	[Nm] Torque amplitude of the selected effect
<i>effect_offset</i>	[Nm] Torque offset of the effect

2. Static inputs: (only used in the launch phase)

<i>SetupX, SetupY, SetupZ</i>	Position of the setup point in the global coordinate system [m]
<i>SetupYaw, SetupPitch, SetupRoll</i>	Rotation of the setup point in the global coordinate system [rad]

3. Static parameters:

<i>CfgDYN</i>	["file.cfg"] base configuration file of the vehicle (section 0 et seq.)
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4. Outputs:

<i>Version</i>	[str] datestring indicating the build date and time
<i>ABS_active</i>	[0..4] Indicates the number of wheels, where ABS is active
<i>RPM</i>	[1/min] Engine rounds per minute
<i>s</i>	[m] traveled distance
<i>Neutral</i>	[0 1] flag to indicate neutral position of automatic transmission
<i>Drive</i>	[0 1] flag to indicate drive position of automatic transmission
<i>Reverse</i>	[0 1] flag to indicate reverse position of automatic transmission
<i>Consumption</i>	[l/100km] instantaneous fuel consumption
<i>Fuel_avg</i>	[l/100km] average fuel consumption (trip)
<i>Fuel_used</i>	[l] total consumed fuel
<i>X, Y, Z</i>	[m] position of vehicle reference point w.r.t earth
<i>roll, pitch, yaw</i>	[rad] vehicle Euler angles
<i>X_, Y_, Z_cog</i>	[m] position of vehicle center of gravity w.r.t earth
<i>v, v_kmh</i>	[m/s km/h] velocity of the vehicle
<i>EngineTorque</i>	[Nm] currently induced drive torque of the engine
<i>EngineOmega</i>	[rad/s] engine speed
<i>EnginePower</i>	[W] currently induced engine power
<i>Eng_Tmax, req, loss</i>	[Nm] currently available maximum engine torque, driver torque request and engine loss torque



SILAB DRIVING SIMULATION

version Version 7.1 documentation

<i>SelGear</i>	[+1...+n] Gear selected by automatic transmission
<i>Xfl, Yfl, _fr, ...</i>	[m] position of tire contact point w.r.t. earth
<i>omega_fl, _fr, ...</i>	[rad/s] wheelspeeds
<i>lambda_fl, _fr, ...</i>	[%] longitudinal wheel slip
<i>alpha_fl, _fr, ...</i>	[rad] sideslip angle
<i>spring_fl, _fr, ...</i>	[m] suspension spring travel
<i>delta_fl, _fr, ...</i>	[rad] wheel individual steering angle
<i>Fz_fl, _fr, ...</i>	[N] wheel load
<i>vx, vy, vz</i>	[m/s] lon / lat / vert velocity of the vehicle
<i>vroll, vpitch, vyaw</i>	[rad/s] angular velocities of the vehicle
<i>ax, ay, az</i>	[m/s ²] lon / lat / vert acceleration of the vehicle
<i>aroll, apitch, ayaw</i>	[rad/s ²] angular accelerations of the vehicle
<i>SteeringMoment</i>	[Nm] Torque at steering wheel resulting from tire forces
<i>effect_torque</i>	[Nm] added effect torque
<i>SteeringVelocity</i>	[rad/s] angular velocity of mockup steering device
<i>road_waves</i>	[m] artificial harmonic road excitation signal
<i>MotionRoll</i>	[rad] platform roll angle of internal motion cueing
<i>MotionPitch</i>	[rad] platform pitch angle of internal motion cueing
<i>StandStill</i>	[0 1] flag to indicate vehicle standstill
<i>PauseTrigger</i>	[0 1] flag to suggest simulation pause (e.g. due to vehicle tilt)

5. Dependencies to other DPUs:

DPUDYNCar can run together with **DPUVDSSCNX** if road height and friction adjustments shall be used. The Index value of **DPUDYNCar** must be higher than **DPUVDSSCNX**'s Index value. **DPUDYNCar** allows for the use the "enable_refinement" property of **DPUVDSSCNX**.

DPUDYNCar provides a steering moment signal that can be used with force feedback devices e.g. by use of the **DPUDX8RacingWheel**. When setting up your steering system, remember to scale between physical values (Nm) and normalized values (-1...+1)! Also ensure, that all threads related somehow with the steering actuation (including **DPUDYNcar**) use commonly high frequencies (120Hz or higher). When you experience noise and vibrations in the steering motor, check the *SteeringVelocity* output of **DPUDYNCar** and adjust the thread frequencies and IDs with care.

Configuration Files:

The vehicle configuration is separated into multiple files, such that various combinations can easily be implemented. The main configuration file that contains the links to the specific component configurations should be located at

lib/configurations/DYN

The *yellow* parameters listed below refer to arrays in the configuration files.



6. Main configuration file:

DYN model {...}; list of includes, linking to the component configuration files

7. SystemConstants configuration file: DYN const {...};

rho_air [kg/m³] density of air
Sys_v_low [m/s] velocity at which stand-still transition is active. This will influence starting and stopping behavior as well as oscillations during standstill, if other parameters are chosen imperfectly.

8. Road configuration file: DYN road {...}; (experimental)

The road definition file allows to implement harmonic excitations of a motion platform. This can positively affect the simulator immersion. This implementation does not affect the vehicle dynamics! A set of four simple sine waves is superponed:

Road_Wave [1|0] activity of single sine wave
 [1x4]-array: {ID1, ID2, ID3, ID4}
Road_AmpZ [m] amplitude of each sine wave
 [1x4]-array: {z1, z2, z3, z4}
Road_LenZ [m] wavelength of each sine wave
 [1x4]-array: {l1, l2, l3, l4}

9. Chassis configuration file: DYN chassis {...};

The vehicle body is simplified as a rigid body with the following geometrical and inertial properties. The reference point is used by the graphical object and may not align with the vehicle's center of gravity.

Vehicle_cWA [-] drag resistance coefficient
Vehicle_cD [-] downforce coefficient
Vehicle_m [kg] mass of body
Vehicle_lxx, lyy, lzz [kgm²] mass moment of inertia of body around x/y/z-axis
Vehicle_h_CoG [m] height of center of gravity (COG)
Vehicle_h_Aero [m] effective height of aerodynamic drag force
Vehicle_x_Aero [m] effective longitudinal position of downforce
Vehicle_Whl [m] position of each wheel w.r.t. COG
 [1x12]-array: {xfl, yfl, zfl, xfr, yfr, zfr, xrl, yrl, zrl, xrr, yrr, zrr}
Vehicle_Ref [m] position of reference point w.r.t. COG
 [1x3]-array: {x_ref, y_ref, z_ref}
Vehicle_p0 [m] initial position of COG w.r.t earth
 [1x3]-array: {x_0, y_0, z_0}

The initial z-position value is reevaluated during startup depending on e.g. tire and suspension stiffness, mass, wheelbase, etc.! Initial x- and y-position should typically be zero. Offsets are preferably dealt with in SetupX, -Y, -Z.



10. Wheel configuration file: DYN wheels {...};

The car body allows to use four wheels. Array values (indicated orange) allow for different parametrization of each wheel. The tire forces are calculated according to Pacejka's Magic Formula.

#Pacejka:

<i>Wheel_Tire_mu_max_4x1</i>	[-] maximum friction coefficient
<i>Wheel_Tire_c_start_4x1</i>	[-] slope of mu-slip-curve in origin
<i>Wheel_Tire_mu_slip_4x1</i>	[-] friction coefficient at full sliding
<i>Wheel_Tire_lambda_mu_max_4x1</i>	[-] slip value at maximum force
<i>Wheel_Tire_mu_Gain_4x1</i>	[-] Gain factor for fast adjustment

#Other:

<i>Wheel_rdyn</i>	[m] dynamic roll radius
<i>Wheel_lyy</i>	[kgm ²], inertia of wheel around y-axis
<i>Wheel_m</i>	[kg] wheel mass
<i>Wheel_c_Tire</i>	[N/m] tire stiffness
<i>Wheel_c_gamma</i>	[N/rad] camber stiffness
<i>Wheel_c_alpha</i>	[N/rad] slip angle stiffness
<i>Wheel_c_lambda</i>	[N/rad] longitudinal slip stiffness
<i>Wheel_relax_length_vertical</i>	[m] relaxation length of vertical force generation
<i>Wheel_relax_length_lateral</i>	[m] relaxation length of horizontal force generation
<i>Wheel_fr_const</i>	[-] roll resistance coefficient, constant
<i>Wheel_fr_sqr</i>	[-] roll resistance coefficient, wheel rotation dependency
<i>Wheel_slip_vmin</i>	[m/s] lower limit velocity for slip calculation
<i>Wheel_c_still</i>	[N/m] horizontal tire stiffness at standstill
<i>Wheel_d_still</i>	[Ns/m] horizontal tire damping at standstill
<i>Wheel_Fx,y,zSat</i>	[N] maximum tire forces per coordinate direction
<i>Wheel_trail</i>	[m] geometric trail of the wheel
<i>Wheel_toe_in</i>	[rad] toe angle
<i>Wheel_camber</i>	[rad] camber angle

11. Brake configuration file: DYN brakes {...};

The maximum braking potential is defined in the configuration file, while the brake pedal input will only be set between 0 and 1. The use of ABS is optional and can be set in the model's input.

<i>Brk_T_ped_1xn</i>	[Nm] maximum brake torque at each wheel
<i>Brk_T_hand_1xn</i>	[Nm] maximum hand-brake torque at each wheel

#ABS:

<i>Brk_Kp</i>	[-] P-Portion of PID-Controller
<i>Brk_Ki</i>	[-] I-Portion of PID-Controller
<i>Brk_lambda_nom</i>	[-] nominal set slip for slip controller
<i>Brk_cp</i>	[m ³] brake-torque-ratio
<i>Brk_Tau</i>	[-] filter constant



12. Engine Configuration File: DYN engine {...};

The engine behavior is modeled by a maximum torque curve and throttle position creates a power demand. Friction losses and auxiliary power demand can be implemented.

#General

Eng_dir	Crankshaft Direction: +-1 = Longitudinal, +-2 = Lateral
Eng_J	[kgm ²] crankshaft inertia
Eng_tau	[s] time lag constant of throttle-torque response

#Torques

Eng_T_min	[Nm] minimum engine torque
Eng_T_max	[Nm] maximum engine torque
Eng_T_n_max	[Nm] Torque at n_max
Eng_T_fric_min	[Nm] minimum friction torque
Eng_T_fric_max	[Nm] maximum friction torque
Eng_T_lim_gear	[0...1] Torque gain k_g per gear
[1x8]-array:	{ $k_{Reverse}$, $k_{Neutral}$, k_{Gear1} , ..., k_{Gear6} }

#RPMs (increasing order!)

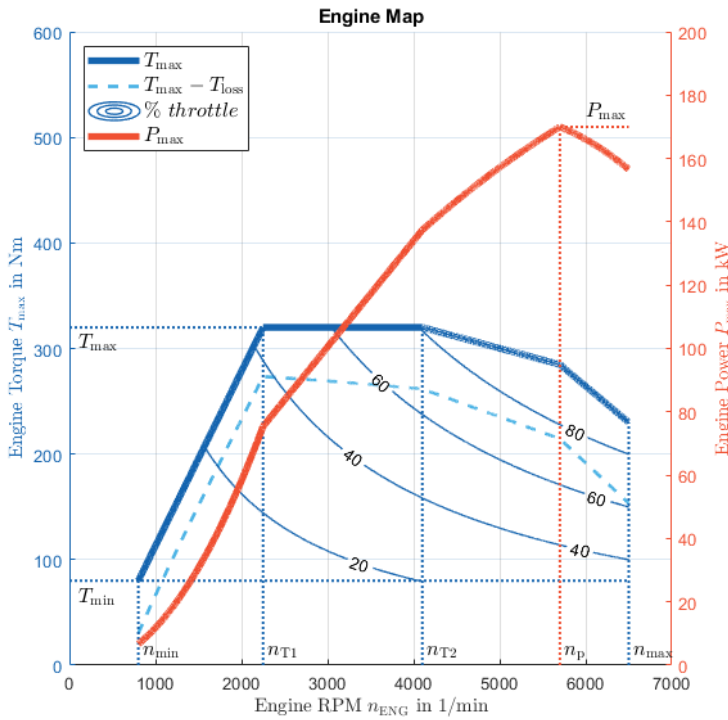
Eng_n_min	[1/min] minimum engine rotation speed
Eng_n_idle	[1/min] idle rotation speed
Eng_n_T1	[1/min] min rotation speed at torque plateau
Eng_n_T2	[1/min] max rotation speed at torque plateau
Eng_n_p	[1/min] rotation speed at maximum engine power
Eng_n_max	[1/min] minimum engine rotation speed

#Power

Eng_P_max	[W] maximum engine power
Eng_P_aux	[W] auxiliary power loss
Eng_Pfric_n_pmax	[W] loss at maximum power

#Fuel Characteristics:

rho_fuel	[kg/l] density of fuel (Diesel: 0.83, Petrol: 0.74)
Eng_fuel_paraboloid	[0 1] when set to 0, the specific consumption is constant
Eng_b_i	[kg/kWh] specific fuel consumption (constant)
Eng_b_i_opt	[kg/kWh] optimum specific fuel consumption
Eng_b_i_max	[kg/kWh] maximum specific fuel consumption
Eng_b_i_Tsqr	[-] quadratic rise factor over normalized engine torque
Eng_b_i_nsqr	[-] quadratic rise factor over normalized engine speed
Eng_T_fuel_opt	[Nm] Torque where the specific fuel consumption is optimal
Eng_n_fuel_opt	[rpm] Engine speed where the specific fuel consumption is optimal



Parameter	Value
Eng_T_min	80
Ent_T_max	320
Eng_T_n_max	230
Eng_T_fric_min	20
Eng_T_fric_max	25
Eng_n_min	800
Eng_n_idle	900
Eng_n_T1	2250
Eng_n_T2	4100
Eng_n_p	5700
Eng_n_max	6500
Eng_P_max	170000
Eng_P_aux	2000
Eng_Pfric_n_pmax	6000

Figure 1: The example engine map shows the four part piecewise linear approximation of the maximum torque per rpm and the resulting maximum power. The contour lines represent torque iso-lines at constant power, given at certain throttle percentages.

13. Drivetrain configuration file: DYN drivetrain {...};

The user may choose between a manual or automatic gearbox.

Gbx_type	[-] 1 = Manual; 2 = Automatic;
Gbx_d_tr	[Nms/rad] transmission damping coefficient
Gbx_eta_tr	[-] efficiency of transmission
Gbx_J	[kgm ²] mass moment of inertia of transmission
Gbx_i_all	[-] array of transmission ratios
[1xn]-array:	{i_1, i_2, i_3, ..., i_n}; max(n) = 16
Gbx_i_rev	[-] transmission ratio of reverse gear
Gbx_nGear	[-] number of gears (without reverse);
Gbx_ShiftDelay	[s] delay until automatic transmission may shift again
Gbx_ThreshPedal	[0...1] pedal threshold needed to allow gear shifts in automatic mode
Gbx_ThreshFT	[0...1] pedal value that indicates "full throttle"
Gbx_ThreshFB	[0...1] pedal value that indicates "full brake"
Gbx_UPrpm	[1/min] trigger rpm for upshifting
Gbx_UPrpm_FT	[1/min] trigger rpm for upshifting at full throttle
Gbx_DOWNrpm	[1/min] trigger rpm for downshifting
Gbx_DOWNrpm_FB	[1/min] trigger rpm for downshifting at full brake



SILAB DRIVING SIMULATION

version Version 7.1 documentation

<i>Gbx_PowerShiftFactor</i>	[-] power demand factor to allow power shift
<i>Gbx_PowerShiftDelay</i>	[s] time delay to allow power shift
	<i>A power shift (down) is executed, when the power demand is greater than the currently available maximum power over a duration longer than the power shift delay.</i>
<i>Gbx_ShiftBackDelay</i>	[s] additional delay for alternating shift sequences

#Torque Converter

The Vehicle model is equipped with a converter rather than a disc clutch. The torque capacity of the converter is proportional to the difference of squared angular velocities of input and output shaft.

<i>TConv_Cap_factor</i>	[Nm/Hz ²] capacity factor of the converter
<i>TConv_eta_max_tc</i>	[-] maximum efficiency at v=0.5

#Differentials

The user may choose between different drive configurations by setting up the differentials accordingly. Front- and rear-wheel drive as well as 50/50 AWD are possible.

<i>Diff_mode</i>	[-] 1=front, 2=rear, 3=AWD
<i>Diff_front_d</i>	[Nms/rad] damping of differential
<i>Diff_front_i_final</i>	[-] gear ratio of final drive
<i>Diff_front_J</i>	[kgm ²] Moment of Inertia kgm ²
<i>Diff_front_eta</i>	[-] efficiency of differential
<i>Diff_front_lock</i>	[0...1] diff lock
<i>Diff_rear_...</i>	see above
<i>Diff_center_...</i>	see above (no diff lock!)



14. Suspension Parameters: DYN suspensions {...};

#Suspension Modules

<i>Susp_brsupang_4x1</i>	[rad] brake support angle
<i>[1x4]-array:</i>	{frontleft, frontright, rearleft, rearright}
<i>Susp_trsupang_4x1</i>	[rad] traction support angle
<i>Susp_latsupang_4x1</i>	[rad] tangent of lateral support angle
<i>Susp_c_antirollbar_front</i>	[Nm/rad] anti-roll-bar stiffness front
<i>Susp_c_antirollbar_rear</i>	[Nm/rad] anti-roll-bar stiffness rear
<i>Susp_c_Body_4x1</i>	[N/m] stiffness
<u>#Damper Setup</u>	
<i>Susp_SimpleDamper</i>	[0 1] use 0 for pure linear damping
<i>Susp_d_Body_4x1</i>	[Ns/m] linear damping constant
<i>Susp_d_off_comp_4x1</i>	[Ns/m] offset of damping constant during compression
<i>Susp_d_v_build_comp_4x1</i>	[Ns/m] buildup velocity to reach full compression offset
<i>Susp_d_off_reb_4x1</i>	[Ns/m] offset of damping constant during rebound
<i>Susp_d_v_build_reb_4x1</i>	[Ns/m] buildup velocity to reach full rebound offset

#Steering Setup

<i>Steer_iSteer</i>	[-] transmission ratio of the steering column
<i>Steer_Backlash</i>	[rad] Backlash of the steering column
<i>Steer_TimeConstant</i>	[s] Time constant of the steering column
<i>Steer_PowerSteerFactorMin</i>	[%] minimum torque support factor
<i>Steer_PowerSteerFactorMax</i>	[%] maximum torque support factor
<i>Steer_SteerTorqueMax</i>	[Nm] maximum output steering torque
<i>Steer_FrictionRamp</i>	[Nm/s] friction torque rate limiter
<i>Steer_P</i>	[Nm/rad] proportional part of steer angle controller
<i>Steer_I</i>	[Nm/rads] integral part of steer angle controller
<i>Steer_D</i>	[Nms/rad] derivative part of steer angle controller
<i>Steer_N</i>	[-] filter constant
<i>Steer_T</i>	[Nm] maximum torque of steer angle controller

15. Example:

This DPU works well using the following script:

```
#####  
##### DYNCar Vehicle Dynamics #####  
#####  
  
DPUDYNCar ~DYNcar  
{  
    Index = 10;  
    Computer = {VDYN};  
    TreeviewGroup = "Fahrdynamik";  
    TreeviewGroup~en = "Vehicle Dynamics";  
    CfgDYN = "Limousine.cfg";  
};
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
#####  
#### Road contact simulation #####  
#####
```

```
DPUVDSSCNX ~hRoad  
{  
  Index = 11;  
  Computer = {VDYN};  
  TreeviewGroup = "Fahrdynamik";  
  TreeviewGroup~en = "Vehicle Dynamics";  
  SlopeLimit = 10.0;  
  EnableRefinement = 1;  
};
```

```
#####  
#### Added interface variables #####  
#####
```

```
VDyn.VARSet      += {SteeringMoment,  
                     Wheelspeed_fl,Wheelspeed_fr,  
                     Wheelspeed_rl,Wheelspeed_rr};
```

```
Mockup.VARSet     += {GearUp ,GearDown,  
                     FFB_Damping, FFB_Friction};
```

```
#####  
#### Vehicle state connections #####  
#####
```

```
Connections = {  
  
  ## Road contact Using DPUVDSSCNX for road height input:  
  ~DYNcar.Xfl      -> ~hRoad.Xfl  
  ,~DYNcar.Yfl      -> ~hRoad.Yfl  
  ,~hRoad.Zfl       -> ~DYNcar.Zfl  
  ,~hRoad.mufl      -> ~DYNcar.mufl  
  ,~DYNcar.Xfr      -> ~hRoad.Xfr  
  ,~DYNcar.Yfr      -> ~hRoad.Yfr  
  ,~hRoad.Zfr       -> ~DYNcar.Zfr  
  ,~hRoad.mufr      -> ~DYNcar.mufr  
  ,~DYNcar.Xrl      -> ~hRoad.Xrl  
  ,~DYNcar.Yrl      -> ~hRoad.Yrl  
  ,~hRoad.Zrl       -> ~DYNcar.Zrl  
  ,~hRoad.murl      -> ~DYNcar.murl
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
,~DYNcar.Xrr          -> ~hRoad.Xrr
,~DYNcar.Yrr          -> ~hRoad.Yrr
,~hRoad.Zrr           -> ~DYNcar.Zrr
,~hRoad.murr          -> ~DYNcar.murr

,VDyn.XOffset         -> ~DYNcar.XOffset
,VDyn.YOffset         -> ~DYNcar.YOffset
,VDyn.ZOffset         -> ~DYNcar.ZOffset
,VDyn.SetupX          -> ~DYNcar.SetupX
,VDyn.SetupY          -> ~DYNcar.SetupY
,VDyn.SetupZ          -> ~DYNcar.SetupZ
,VDyn.SetupYaw        -> ~DYNcar.SetupYaw
,VDyn.SetupPitch      -> ~DYNcar.SetupPitch
,VDyn.SetupRoll       -> ~DYNcar.SetupRoll
,VDyn.SetupDone       -> ~DYNcar.SetupDone
,VDyn.SetupDone       -> ~hRoad.SetupDone

## Controls
,VDyn.Ignition         -> ~DYNcar.Ignition
,VDyn.Starter          -> ~DYNcar.Starter
,VDyn.HandBrake        -> ~DYNcar.HandBrake
,VDyn.AcceleratorPedal -> ~DYNcar.AcceleratorPedal
,VDyn.BrakePedal       -> ~DYNcar.BrakePedal
,VDyn.ClutchPedal      -> ~DYNcar.ClutchPedal

## Steering
,VDyn.SteeringWheel    -> ~DYNcar.SteeringWheel
,Mockup.FFB_Damping    -> ~DYNcar.FFB_Damping
,Mockup.FFB_Friction   -> ~DYNcar.FFB_Friction
,~DYNcar.SteeringMoment -> VDyn.SteeringMoment

## Drivetrain
,VDyn.Gear             -> ~DYNcar.GearAutomatic
#,VDyn.Gear            -> ~DYNcar.GearManual
,~DYNcar.Consumption   -> VDyn.Consumption
,~DYNcar.SelGear       -> VDyn.GearAuto
,~DYNcar.RPM           -> VDyn.RPM

## Vehicle States
,~DYNcar.s             -> VDyn.s
,~DYNcar.X             -> VDyn.X
,~DYNcar.Y             -> VDyn.Y
,~DYNcar.Z             -> VDyn.Z
,~DYNcar.yaw           -> VDyn.yaw
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
,~DYNcar.pitch      -> VDyn.pitch
,~DYNcar.roll       -> VDyn.roll
,~DYNcar.v          -> VDyn.v
,~DYNcar.v_kmh      -> VDyn.v_kmh
,~DYNcar.vx         -> VDyn.vx
,~DYNcar.vy         -> VDyn.vy
,~DYNcar.vz         -> VDyn.vz
,~DYNcar.vyaw       -> VDyn.vyaw
,~DYNcar.vpitch     -> VDyn.vpitch
,~DYNcar.vroll      -> VDyn.vroll
,~DYNcar.ax         -> VDyn.ax
,~DYNcar.ay         -> VDyn.ay
,~DYNcar.az         -> VDyn.az
,~DYNcar.ayaw       -> VDyn.ayaw
,~DYNcar.apitch     -> VDyn.apitch
,~DYNcar.aroll      -> VDyn.aroll
,~DYNcar.delta_fl   -> VDyn.delta
,~DYNcar.omega_fl   -> VDyn.Wheelspeed_fl
,~DYNcar.omega_fr   -> VDyn.Wheelspeed_fr
,~DYNcar.omega_rl   -> VDyn.Wheelspeed_rl
,~DYNcar.omega_rr   -> VDyn.Wheelspeed_rr
```

Motion States

```
,~DYNcar.X          -> VDynMotion.X
,~DYNcar.Y          -> VDynMotion.Y
,~DYNcar.Z          -> VDynMotion.Z
,~DYNcar.yaw        -> VDynMotion.yaw
,~DYNcar.pitch      -> VDynMotion.pitch
,~DYNcar.roll       -> VDynMotion.roll
,~DYNcar.v          -> VDynMotion.v
,~DYNcar.vx         -> VDynMotion.vx
,~DYNcar.vy         -> VDynMotion.vy
,~DYNcar.vz         -> VDynMotion.vz
,~DYNcar.vyaw       -> VDynMotion.vyaw
,~DYNcar.vpitch     -> VDynMotion.vpitch
,~DYNcar.vroll      -> VDynMotion.vroll
,~DYNcar.ax         -> VDynMotion.ax
,~DYNcar.ay         -> VDynMotion.ay
,~DYNcar.az         -> VDynMotion.az
,~DYNcar.ayaw       -> VDynMotion.ayaw
,~DYNcar.apitch     -> VDynMotion.apitch
,~DYNcar.aroll      -> VDynMotion.aroll
```

```
};
```